Commit or Recover

A Temporal Safety Monitor for TransFuser++

Course: Intelligent Autonomous Systems	**Primary track:** Safety and Evaluation
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Abstract

This project investigates whether short-term changes in an end-to-end driving policy's intended motion can warn of failure before a reactive safety check. A pretrained TransFuser++ agent will remain frozen in CARLA while a lightweight GRU monitors its recent spatial paths, target-speed probability distributions, controller outputs, ego motion, and route context. The monitor predicts collision, off-road, or route-deviation risk within a fixed horizon and invokes a common safe-stop fallback. We will compare it with current-path geometry, native TF++ target-speed confidence, and handcrafted temporal rules under matched intervention budgets. The central outcome is a closed-loop safety-utility comparison: failures prevented versus unnecessary interventions and lost route progress.

1. Problem Formulation

1.1 Motivation

End-to-end driving systems can fail near discrete decision points such as roundabout exits, lane splits, merges, and obstacle detours. A policy may alternate between incompatible paths, change target speed abruptly, or issue rapidly reversing controls before a current-path collision check recognizes an immediate hazard. If execution continues, a correctable planning instability can become a collision, off-road event, or route deviation. The challenge is to detect dangerous inconsistency without treating every legitimate correction as a reason to stop.

We therefore study the evolving output behavior of a strong frozen policy rather than building another driving model. This makes the central comparison observable, reproducible, and narrow enough for a semester project.

RESEARCH QUESTION

Does temporal inconsistency in TransFuser++'s predicted paths, target-speed distributions, and control outputs provide earlier and more selective warning of closed-loop driving failures than instantaneous geometric checks or single-frame confidence measures?

1.2 Hypotheses

Hypothesis	Testable prediction
H1 - Detection	A GRU using 1-3 s of TF++ output history will improve recall and warning time over current-frame geometry and target-speed confidence at the same intervention budget.
H2 - Closed-loop value	Earlier temporal warnings will prevent more baseline failures with less route-completion loss than reactive monitoring.
H3 - Signal value	Path evolution, speed-distribution change, and controls will contribute complementary predictive information.

SCOPE BOUNDARY

The project does not retrain TF++, learn a general recovery planner, or claim that runtime monitoring is new. It isolates whether TF++-specific temporal instability adds value when policy, scenarios, fallback, and intervention rate are held fixed.

2. Proposed Technical Approach

2.1 Runtime architecture

A single released TF++ checkpoint runs as the policy under test. TF++ separates lateral and longitudinal intent: it predicts a distance-sampled spatial path and a probability distribution over target-speed classes; its downstream controller converts them to steering, throttle, and brake [2]. The monitor observes these outputs over time without changing the policy.

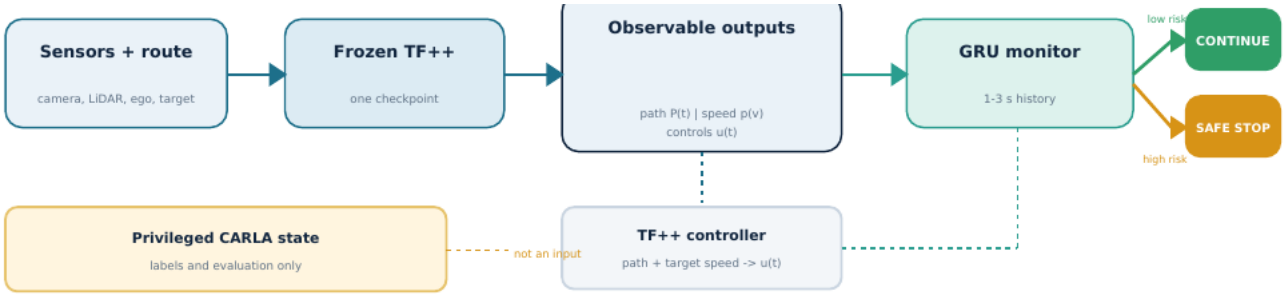


Figure 1. The monitor observes external TF++ outputs and non-privileged context; every monitor invokes the same fallback.

MONITOR STATE
At time t, z_t = [P_t, p_t(v), u_t, s_t, r_t], where P is the spatial path, p(v) the target-speed distribution, u the controls, s ego motion, and r route/target information. A GRU maps the 1-3 s history to the probability of failure within the next 3 s.

2.2 Operational definition of instability

Signal family	Observable indicators
Path evolution	Mean pointwise disagreement after ego-motion alignment; endpoint and heading variance; curvature change; maneuver-mode switching.
Speed distribution	Entropy, top-class confidence margin, and Jensen-Shannon divergence between consecutive target-speed distributions.
Controls	Steering reversals, total action variation, steering rate, throttle/brake switching, and longitudinal jerk.
Execution context	Mismatch between intended path and realized motion; alignment with route target; short-term route-progress consistency.

Instability is not automatically unsafe. Sharp corrections and plan changes may be appropriate. Training data will therefore include successful turns and benign replanning as hard negatives so the monitor must learn instability plus consequence, not merely large steering magnitude.

2.3 Monitor training and intervention

Element	Design choice
Prediction	Binary risk within 3 s, trained from future CARLA outcomes with class weighting or balanced sampling.
Calibration	Choose the alert threshold only on validation scenarios to meet the common intervention budget.
Fallback	Release throttle and apply bounded braking while maintaining stable lateral control; no learned replanning in the minimum viable experiment.
Runtime	Log monitor latency and keep the GRU small enough to run alongside TF++ without changing its control rate.

3. Experimental Design

3.1 Fixed implementation stack

Component	Choice	Experimental role
Simulator	CARLA 0.9.15	Deterministic scenario execution and privileged labels.
Policy	TF++ from CARLA Garage, leaderboard_2	Frozen single checkpoint in the primary experiment.
Evaluation routes	Selected Bench2Drive routes + custom cases	Short, repeatable decision-point trials.
Monitor	Small GRU	Predicts imminent failure from 1-3 s of history.
Fallback	Bounded safe stop	Identical across monitored conditions.

3.2 Scenario and data generation

The full Bench2Drive training set is unnecessary because TF++ remains frozen. We will use the repository's closed-loop tooling, select relevant Bench2Drive scenario families [7], and add parameterized cases emphasizing discrete commitment points. Each family varies geometry, speed, traffic seed, weather, route target, and obstacle placement.

Scenario family	Controlled difficulty	Observed outcome
Roundabout exit	Exit choice, approach angle, lane position	Median contact, wrong exit, late turn
Lane split / merge	Late route commitment and competing paths	Oscillation, route deviation, off-road
Obstacle detour	Two plausible paths around obstruction	Collision, blockage, repeated path switching
Narrowing / occlusion	Temporary boundary or uncertain free space	Curb contact, unsafe continuation, false stop

3.3 Labels and leakage-safe splits

- A frame is positive when unchanged TF++ causes a collision, off-road violation, severe route deviation, or unrecoverable blockage within the next 3 s; a 2 s horizon is an ablation.
- Privileged simulator state supplies labels and metrics only. It is never an input to the primary monitor.
- All frames from one scenario seed remain in one split. Generalization is tested with held-out scenario families, configurations, traffic seeds, and weather - not by claiming town-level novelty from an all-town checkpoint.

3.4 Primary comparison

Condition	Monitor signal	Purpose
No monitor	None	Original TF++ failure and utility rate.
Geometric	Current path + obstacles/drivable area	Reactive physical-risk baseline.
TF++ confidence	Current target-speed entropy or margin	Tests whether native single-frame confidence is sufficient.
Handcrafted temporal	Path disagreement, steering flips, speed-distribution change	Tests whether fixed temporal rules are sufficient.
Learned temporal	Full path, speed, control, ego, and route history	Proposed GRU monitor.

MINIMUM VIABLE EXPERIMENT

One frozen TF++ checkpoint, three core scenario families, one safe-stop controller, five monitor conditions, a 3 s failure horizon, and held-out paired closed-loop evaluation with three GRU seeds.

3.5 Logged unit of analysis

Record group	Fields
Policy outputs	Timestamped spatial path, full target-speed distribution, and steering/throttle/brake after control conversion.
Context	Ego speed, acceleration, yaw rate, route target, scenario family, configuration, traffic seed, and weather.
Outcome	Infraction type, time-to-failure, route progress, minimum obstacle distance, and whether fallback was invoked.
Traceability	TF++ checkpoint, monitor version, threshold, random seed, and paired scenario identifier.

4. Evaluation and Analysis Plan

4.1 Metrics

Metric group	Measures
Early warning	AUPRC; recall at fixed false-positive/intervention rate; mean warning time; recall at least 1 s and 2 s before failure.
Closed-loop safety	Collision and off-road rate; route deviation; minimum obstacle distance; fraction of baseline failures prevented.
Utility	Route completion; driving score; unnecessary interventions; stopped time; travel-time increase.
Comfort and cost	Harsh braking; longitudinal/lateral jerk; monitor latency and compute overhead.

4.2 Controlled closed-loop protocol

- Run each held-out scenario seed first with unmodified TF++, then rerun it with every monitor under identical environment parameters.
- Use the same safe-stop behavior for all monitored methods so fallback quality cannot explain detector rankings.
- Select thresholds on validation data at a matched intervention budget; also report safety-utility curves across thresholds.
- Train the final GRU with three random seeds and report mean +/- standard deviation. Use scenario-level bootstrap intervals and paired success/failure transitions where practical.

4.3 Ablations and generalization

Axis	Comparison	Question
Temporal context	Current frame vs 1, 2, and 3 s history	Is history itself useful?
Input groups	Path only; speed only; controls only; path + speed; all inputs	Which TF++ signals matter?
Representation	Raw sequences vs handcrafted instability features	Does learned temporal structure add value?
Data balance	With vs without benign-correction hard negatives	Does the model avoid reflexive stopping?
Generalization	Held-out scenario family and severity ranges	Does the signal transfer beyond one case type?
Policy variation - stretch	Other released TF++ training seeds	Is the monitor tied to one checkpoint's quirks?

SUCCESS CRITERION

At the same intervention rate, the learned temporal monitor must prevent more failures than both the geometric and TF++-confidence baselines, while avoiding a disproportionate loss in route completion. Classification accuracy alone is not sufficient.

4.4 Statistical decision rules

Claim	Evidence required
Predictive value	Learned GRU exceeds geometry and TF++ confidence in AUPRC and recall at the common budget.
Timing value	Alerts precede the original failure with enough margin for the fixed stop; report the full warning-time distribution.
Safety value	Paired reruns convert more baseline failures to successes than they convert baseline successes to disrupted runs.
Utility value	The safety-utility curve dominates or improves the strongest baseline over a meaningful threshold range.

4.5 Interpretation of outcomes

A positive result would show that cross-time output inconsistency is actionable beyond TF++'s instantaneous confidence. A negative result remains useful: geometry or native confidence may already capture the relevant risk, or observable outputs may be insufficient without internal policy features.

5. Relation to Recent Work and Novelty

Recent work already establishes learned runtime monitoring, temporal safety forecasting, and online repair. The defensible gap is therefore deliberately narrow: does temporal inconsistency in a frozen TF++ policy add warning beyond current-frame geometry and TF++'s own confidence when detection is isolated from recovery?

Work	Year	Relevant contribution	Boundary from this project
[1] EvoEye	2026	Temporal fusion of heterogeneous ADS messages plus evolving scenario acquisition.	Closest temporal monitor; ours isolates TF++ output signals without a self-evolving loop.
[2] TF++	2023	Spatial path and target-speed classification for end-to-end driving.	Frozen policy under test and source of the confidence baseline.
[3] MarMot	2023/24	Metamorphic consistency across transformed current inputs.	Black-box consistency baseline, but not natural cross-time plan instability.
[4] UQ warning	2024	MC Dropout and ensembles predict lane-keeping misbehavior.	Supports warning-time metrics; our MVE uses TF++'s native confidence.
[5] Temporal metrics	2024	Forecasts future safety metrics from outputs and context.	Closest conceptual predecessor; ours focuses on decision-point TF++ instability.
[6] Validity learning	2024	Learns from rollout failures to improve trajectory validity.	Modifies planning offline; ours monitors a frozen policy online.
[7] Bench2Drive	2024	Short, multi-ability closed-loop CARLA evaluation.	Scenario and metric resource, not a runtime detector.
[8] EnDfuser	2025/26	Diffusion trajectory samples expose within-frame uncertainty.	Conceptual comparison to our cross-time deterministic-path inconsistency.
[9] ADReFT	2025	Joint state monitor and adaptive decision repair.	Direct novelty risk; our fixed fallback isolates detection signal quality.
[10] Argus	2025	Trajectory hazard monitor with IDM-based mitigation.	Strong system competitor; ours targets warnings before current-path hazard checks.

NOVELTY STATEMENT

We do not modify or retrain TransFuser++. We test whether residual temporal inconsistency in its externally observable path, target-speed, and control outputs predicts imminent decision-point failures beyond current-frame geometric risk and target-speed confidence, while policy, fallback, scenarios, and intervention budget remain fixed.

What this project claims	What it does not claim
A controlled TF++-specific signal study; paired closed-loop safety-utility evidence; and a reproducible decision-point benchmark.	The first runtime monitor or repair system; a new end-to-end driving policy; or production-readiness and real-road validation.

5.1 Evidence chain for the final claim

Stage	Required finding
1 - Signal	Temporal TF++ history predicts future failure beyond current-frame geometry and confidence.
2 - Timing	Its additional detections occur early enough for the fixed fallback to change the outcome.
3 - Closed-loop value	Those interventions reduce infractions without an unacceptable route-progress or false-stop penalty.

6. Deliverables, Timeline, Risks, and References

6.1 Expected contributions

- A reproducible decision-instability benchmark with parameterized CARLA cases, TF++ histories, future-failure labels, and leakage-safe splits.
- A lightweight GRU monitor and a controlled comparison against geometry, native TF++ confidence, and handcrafted temporal instability.
- Paired closed-loop safety-utility results showing both prevented failures and the cost of unnecessary intervention.
- An ablation study identifying whether path, target-speed, controls, ego dynamics, and route context provide distinct value.

6.2 Primary risks and mitigations

Risk	Mitigation
Too few failures	Systematically vary decision geometry, speed, route goal, traffic seed, and bounded sensor noise.
Trivial imminent-collision labels	Forecast 3 s ahead and compare with a strong current-path geometric check.
Always-stop solution	Match intervention budgets; report route completion, delay, false stops, and comfort.
Monitor learns steering magnitude	Include sharp successful turns and benign plan changes as hard negatives.
Infrastructure risk	Freeze one checkpoint; prioritize three scenario families and fixed safe stop; gate transfer as stretch work.

6.3 References

[1] M. Cheng, L. Briand, and X. Xie. EvoEye: Self-Evolving Runtime Monitoring for Autonomous Driving Systems. arXiv:2607.03755, 2026.

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